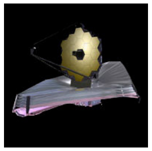

11856 - Does the wide-separation lensed quasar SDSS J1326+4806 host a dual AGN?

Cycle: 5, Proposal Category: GO

INVESTIGATORS

Name	*Institution*
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Adi Foord (CoI)	University of Maryland Baltimore County
Dr. Michael Florian (CoI)	Eureka Scientific Inc.

OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
NIRSpec IFU Prism				
	1	Prism	NIRSpec IFU Spectroscopy	(1) SDSSJ1326+4806-A
	2	Prism Background	NIRSpec IFU Spectroscopy	(3) SDSSJ1326+4806-B
NIRSpec IFU Grating				
	4	Grating Spectroscopy	NIRSpec IFU Spectroscopy	(1) SDSSJ1326+4806-A
NIRCam				
	3	Repeat Imaging one to check for variability	NIRCam Imaging	(2) SDSSJ1326+4806-Cluster
	6	Repeat Imaging two to check for variability	NIRCam Imaging	(2) SDSSJ1326+4806-Cluster

ABSTRACT

The wide separation lensed quasar, SDSS J1326+4806 (z=2.08), was recently imaged by JWST and HST. Its host, which in ground-based observations is dominated by the light of the quasar, is revealed in the high-resolution space data to be a resolved face-on spiral galaxy. A second compact source is visible in both images of the lensed quasar host, in projected distance of only 1.5 kpc from the central quasar. This program aims to obtain spectroscopy of the host galaxy, to determine whether the second source is an active galactic nucleus (AGN) and quantify its variability. If confirmed, this would be the closest-separation dual AGN known at z>0.5 -- two supermassive black holes caught just before they become gravitationally bound and coalesce into a single black hole through emission of low frequency gravitational waves. The data collected in this program will lead to a comprehensive study of the host galaxy at sub-kpc scales, benefitting from the magnification boost of strong gravitational lensing.

OBSERVING DESCRIPTION

1) NIRSpec IFU Prism spectroscopy to identify and characterize the candidate 2nd AGN; The wavelength coverage is 1950A-17219A, spans the range of many lines including key broad lines for mass estimates from scaling relations.
Because the target completely fills the IFU, we include a dedicated background observation separated by ~8 arcseconds from the main target.

2) NIRSpec IFU grating spectroscopy with high resolution gratings (100km/sec resolution) will be used for studying the dynamics within the host galaxy. The wavelength window includes key lines, H-beta through SII; we will use two configurations, G140H/F100LP + G235H/F170LP.

3) New imaging in NIRCam/F090W (1073 sec) in two epochs will be paired with concurrent imaging in F277W (1073 sec) and F356W (1073 sec). Two epochs of NIRCam imaging in F115W (601 sec), F150W (429 sec) and F444W (1030 sec) will provide a second and third imaging epoch to the observations from GO-6675. The purpose of this duplication is to check for and quantify the variability of both components of the candidate dual AGN -- the known central quasar, and the new candidate.

*GO-6675 that this program partially duplicates is a joint HST+JWST program to observe several lensed quasars, with a science goal of measuring the Hubble constant from time delays. This field was observed in Feb 2025.

Proposal 11856 - Targets - Does the wide-separation lensed quasar SDSS J1326+4806 host a dual AGN?

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	SDSSJ1326+4806-A	RA: 13 26 0.1788 (201.5007450d) Dec: +48 06 37.50 (48.11042d) Equinox: J2000	Parallax: 0"	
Comments: Category=Galaxy Description=[Active galactic nuclei, Quasars, Spiral galaxies] Extended=YES				
(2)	SDSSJ1326+4806-Cluster	RA: 13 25 59.5580 (201.4981583d) Dec: +48 06 49.37 (48.11371d) Equinox: J2000	Parallax: 0"	
Comments: Strong lensing Category=Clusters of Galaxies Description=[Brightest cluster galaxies, Rich clusters] Extended=YES				
(3)	SDSSJ1326+4806-B	RA: 13 26 0.8755 (201.5036479d) Dec: +48 06 34.06 (48.10946d) Equinox: J2000		
Comments: Background field as primary target completely fills the IUF Category=Galaxy Description=[Active galactic nuclei, Quasars, Spiral galaxies]				

Observation

Proposal 11856, Observation 1: Prism

Fri Mar 13 22:03:16 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	SDSSJ1326+4806-A	RA: 13 26 0.1788 (201.5007450d) Dec: +48 06 37.50 (48.11042d) Equinox: J2000	Parallax: 0"	

Comments:
Category=Galaxy
Description=[Active galactic nuclei, Quasars, Spiral galaxies]
Extended=YES

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	CYCLING	SMALL	1	6	

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	NRSIRS2RAPID	10	1	false	true	NONE	6	6	962.867	277554.1

Special Requirements

Group Observations 1, 2, Non-interruptible

Observation

Proposal 11856, Observation 2: Prism Background

Fri Mar 13 22:03:16 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	SDSSJ1326+4806-B	RA: 13 26 0.8755 (201.5036479d) Dec: +48 06 34.06 (48.10946d) Equinox: J2000		

Comments: Background field as primary target completely fills the IUF
Category=Galaxy
Description=[Active galactic nuclei, Quasars, Spiral galaxies]

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	CYCLING	SMALL	1	6	

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	NRSIRS2RAPID	10	1	false	true	NONE	6	6	962.867	

Special Requirements

Group Observations 1, 2, Non-interruptible

Proposal 11856 - Observation 4 - Does the wide-separation lensed quasar SDSS J1326+4806 host a dual AGN?

Observation

Proposal 11856, Observation 4: Grating Spectroscopy

Fri Mar 13 22:03:16 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	SDSSJ1326+4806-A	RA: 13 26 0.1788 (201.5007450d) Dec: +48 06 37.50 (48.11042d) Equinox: J2000	Parallax: 0"	

Comments:
Category=Galaxy
Description=[Active galactic nuclei, Quasars, Spiral galaxies]
Extended=YES

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	CYCLING	SMALL	1	9	

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G140H/F100LP	NRSIRS2RAPID	20	1	false	true	NONE	9	9	2757.3	277554.2
2	G235H/F170LP	NRSIRS2RAPID	20	1	false	true	NONE	9	9	2757.3	277554.3

Observation

Proposal 11856, Observation 3: Repeat Imaging one to check for variability

Fri Mar 13 22:03:16 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	SDSSJ1326+4806-Cluster	RA: 13 25 59.5580 (201.4981583d) Dec: +48 06 49.37 (48.11371d) Equinox: J2000	Parallax: 0"	

Comments: Strong lensing
Category=Clusters of Galaxies
Description=[Brightest cluster galaxies, Rich clusters]
Extended=YES

Template

Module	Subarray	Target Placement
ALL	FULL	Module B center (small extended source)

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	INTRAMODULEBOX	4	STANDARD		1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F115W	F444W	BRIGHT2	7	1	4	4	601.259	143577
2	F150W	F444W	BRIGHT2	5	1	4	4	429.471	143577
3	F090W	F277W	BRIGHT2	6	2	8	4	1073.677	

Special Requirements

Offset -30.0 arcsec, -30.0 arcsec
Fiducial Point Override NRCBS_FULL

6 After 3 by 60 Days to <None specified>

Observation

Proposal 11856, Observation 6: Repeat Imaging two to check for variability

Fri Mar 13 22:03:16 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	SDSSJ1326+4806-Cluster	RA: 13 25 59.5580 (201.4981583d) Dec: +48 06 49.37 (48.11371d) Equinox: J2000	Parallax: 0"	

Comments: Strong lensing
Category=Clusters of Galaxies
Description=[Brightest cluster galaxies, Rich clusters]
Extended=YES

Template

Module	Subarray	Target Placement
ALL	FULL	Module B center (small extended source)

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	INTRAMODULEBOX	4	STANDARD		1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F115W	F444W	BRIGHT2	7	1	4	4	601.259	143577
2	F150W	F444W	BRIGHT2	5	1	4	4	429.471	143577
3	F090W	F356W	BRIGHT2	6	2	8	4	1073.677	

Special Requirements

Offset -30.0 arcsec, -30.0 arcsec
Fiducial Point Override NRCBS_FULL

6 After 3 by 60 Days to <None specified>